The lecture hall seated 250 and was filled to overflowing as Isaac Khalatnikov rose to speak. It was a warm summer day in 1965, and the world's leading relativity researchers had gathered in London for the Third International Conference on General Relativity and Gravitation. This was the first opportunity, at such a worldwide gathering, for Isaac Khalatnikov and Evgeny Lifshitz to present the details of their proof that black holes do not contain singularities.

Permission to travel beyond the iron curtain was granted and withdrawn with relative capriciousness in the Soviet Union during the decades between Stalin's death and the Gorbachev era. Lifshitz, though Jewish, had traveled rather freely in the late 1950s, but he was now on a travel blacklist and would remain so until 1976. Khalatnikov had two strikes against him; he was Jewish, and he had never yet traveled abroad. (Permission for one's first trip was exceedingly difficult to win.) However, after a vigorous struggle, including a telephone call in his behalf from the vice-president of the Academy of Sciences, Nikolai Nikolaievich Semenov, to the Central Committee of the Communist party, Khalatnikov had finally won permission to come to London.

As he spoke in the packed London lecture hall, dragging a microphone with him, Khalatnikov wrote equations all over the blackboard, which extended the entire 50-foot width of the room. His were not topological methods; they were the standard, equation-intensive methods that physicists had used for decades when analyzing spacetime curvature. Khalatnikov demonstrated mathematically that random perturbations must grow as a star implodes. This meant, he asserted, that if the implosion is to form a singularity, it must be one with completely random deformations in its spacetime curvature. He then described how he and Lifshitz had searched, among all types of singularities permitted by the laws of general relativity, for one with completely random curvature deformations. He exhibited, mathematically, one type of singularity after another; he cataloged the types of singularities almost *ad nauseum*. Among them, none had completely random deformations. Therefore, he concluded—bringing his fortyminute lecture to a close—an imploding star with random perturbations cannot produce a singularity. The perturbations must save the star from destruction.

As the applause ended, Charles Misner, one of Wheeler's most brilliant former students, leaped up and objected strenuously. Excitedly, vigorously,